

Sneha Yenduri

snehayenduri08@gmail.com | 857-206-9037 | United States | <https://www.linkedin.com/in/sneha-yenduri/>
<https://github.com/snehayenduri> | <https://snehayenduri.github.io/>

SUMMARY

Software Engineer with 3+ years of experience building backend-focused full-stack systems with AI/LLM integrations (MCP, OpenAI, LangChain), distributed microservices, and system integrations, deployed on AWS, GCP using Terraform.

WORK EXPERIENCE

Research Software Engineer | Children's Hospital of Philadelphia | USA Feb 2026 – Present

- Engineered a synchronized, high-throughput data acquisition pipeline integrating camera SDKs, NI DAQ, and LED controllers across heterogeneous hardware, reducing end-to-end latency by **20%** and improving signal reliability
- Refactored **MATLAB** codebase to **Python** with **GitHub Copilot & Claude**, achieving 1.4x faster execution and maintainability
- Troubleshoot and optimized data pipelines, improving output reliability by **30%** and enhancing overall system performance

Software Engineer Intern | Oracle (OCI) | California, USA Jan 2025 – Jul 2025

- Built a machine-learning-based real-time log monitoring system with **AI-assisted development (GPT)** that autonomously detected anomalies across large-scale OCI infrastructure, reducing manual investigation effort and speeding up root cause analysis by **5x**
- Tuned and trained the anomaly detection model on **3M+ log lines**, running predictions across **800K+ entries** - boosting detection accuracy by **18%** and cutting false alerts by **30%**, validated through end-to-end integration testing
- Pioneered an MCP-based LLM root cause analysis agent at OCI when the technology had minimal adoption - architecting prompt workflows and orchestration logic, influencing HeatWave LLM agent development and boosting **developer productivity by 35%**
- Architected an async Python full-stack app (**FastAPI, MySQL, Streamlit**) built for high-volume log processing at enterprise scale
- Presented findings and system architecture to **multiple engineering teams** across OCI, driving cross-team visibility and adoption

Senior Software Engineer | Citibank (Fintech) | Pune, India Aug 2021 – Aug 2023

- Designed and delivered **scalable** KYC and financial workflow systems serving multiple global regions, handling enterprise-scale transaction volumes using **Java, Spring Boot, SQL, Microservices, and REST APIs**
- Reduced batch processing time by **35%** via **CI/CD (Jenkins, OpenShift)** and improved code quality by **17%** through JUnit testing
- Championed agile test-driven development practices, improving development speed and system stability by **20%** across the team
- Diagnosed and resolved production and batch workflow failures across four global regions, reducing operational downtime by **4%**

TECHNICAL SKILLS

Programming Languages: Java, C, C++, Python, Shell Scripting, JavaScript, TypeScript, SQL (MySQL, Postgres), NoSQL (MongoDB)

Backend: FastAPI, Flask, Spring Boot, Hibernate, Microservices, RESTful API, Node.js

Frontend: Streamlit, React, Angular, HTML, CSS/SASS, Redux

AI/LLM Integration & Orchestration: MCP (Model Context Protocol), OpenAI APIs, LangChain

Cloud & DevOps: Continuous Integration/Continuous Deployment, Terraform, Packer, Networking Protocols

Tools: Google Cloud Platform, AWS (EC2, S3, IAM, RDS), GitHub, Jenkins, OpenShift, Docker, Tomcat, Postman, SonarQube, Jira, JUnit

Certification: An Oracle Certified Professional (Java SE 11 Developer)

ACADEMIC PROJECTS

MediAssist AI Chatbot | Python, Flask, Streamlit, Pinecone, OpenAI API, LangChain, Hugging Face, AWS, Docker

- Developed an AI-powered medical Q&A system using RAG to ground GPT-4o responses in verified medical literature, reducing hallucinations via context-constrained generation, containerized with Docker and deployed on AWS EC2 via ECR with automated CI/CD through GitHub Actions

Cloud-Ops Automation Platform | GCP, Terraform, Packer, GitHub Actions (YAML workflows), DNS, Node.js

- Architected a cloud-native Node.js backend with Sequelize ORM and CloudSQL, provisioning scalable infrastructure on GCP using Terraform and Packer, implementing Pub/Sub-triggered serverless workflows, and integrating CI/CD pipelines with GitHub Actions alongside VPC networking, load balancing, and autoscaling to achieve 99.99% availability

Project Prism | Java, Angular, Oracle SQL, Spring Boot, Hibernate, Rest API, Amazon EC2

- Engineered a multi-role project management platform with role-based access control, Kanban workflow tracking, and a centralized dashboard for end-to-end project visibility and deployed it in Amazon EC2

EDUCATION

Master of Science in Information Systems | Northeastern University | Boston, USA Sep 2023 - Dec 2025

- Courses:** Program Structures and Algorithms, Data Management and Database Design, Cloud Computing, Web Designing
- Graduating Teaching Assistant** (Network Structures and Cloud Computing): Mentored 50+ students in Cloud (AWS) and DevOps through tutorials and code reviews, emphasizing Linux, Shell Scripting, and Terraform (IaC)